

1. Show that (assuming p is strictly positive), that all eigenvalues of L for Neumann boundary conditions must be non-positive

Solution:

Proof. Start with $\mathcal{L}u = \lambda u$ and multiply by u before integrating to see

$$\begin{aligned}\int_a^b \mathcal{L}u \cdot u \, dx &= \int_a^b \lambda |u|^2 \, dx \Rightarrow \int_a^b (pu')' u \, dx = \lambda \int_a^b |u|^2 \, dx \\ &\Rightarrow pu'u \Big|_a^b - \int_a^b pu' \cdot u' \, dx = \lambda \int_a^b |u|^2 \, dx \\ &\Rightarrow - \int_a^b p |u'|^2 \, dx = \lambda \int_a^b |u|^2 \, dx.\end{aligned}$$

It that $\lambda \leq 0$ since $p > 0$ and u is nontrivial. $\square$

2. Show that at any critical point $x_0 \in [a, b]$ of u_λ , $u_\lambda(x_0) \neq 0$, $u_\lambda''(x_0) \neq 0$, and $u_\lambda''(x_0)$ has the opposite sign to $u_\lambda(x_0)$.

Solution:

Proof. If $u(x_0) = 0$ and $u'(x_0) = 0$ and u solves $\mathcal{L}u = \lambda u$, then u must be the unique solution. Thus, since $u_\lambda(x_0)$ and $u_\lambda''(x_0)$ are non trivial, they cannot be zero.

At a critical point x_0 , we have $u'(x_0) = 0$ so equation (0.3) becomes

$$\begin{aligned}u''(x_0) + \frac{p'}{p}(x_0) \cdot 0 - \frac{\lambda}{p(x_0)} u(x_0) &= \\ \Rightarrow u''(x_0) &= \frac{\lambda}{p(x_0)} u_\lambda(x_0)\end{aligned}$$

Since $\lambda < 0$ (we assume $\lambda \neq 0$) and $p > 0$, the term $\lambda/p(x_0) < 0$. It follows that $u''(x_0) \neq 0$ because $u_\lambda(x_0) \neq 0$. It is also clear that $u_\lambda''(x_0)$ has the opposite sign of $u_\lambda(x_0)$. $\square$

3. Show that the restriction $u_\lambda|_{[x_1, x_2]}$ of u_λ to the subinterval $[x_1, x_2] \subset [a, b]$ is an eigenfunction of $\mathcal{L}$ with Neumann boundary conditions, and

$$\int_{x_1}^{x_2} u_\lambda(x) \, dx = 0.$$

Then use this to show that between every pair of critical points of u_λ , there is a zero of u_λ .

Solution:

Proof. Since x_1 and x_2 are critical points it follows that

$$u_\lambda'(x_1) = u_\lambda'(x_2) = 0$$

which is exactly the Neumann boundary conditions on $[x_1, x_2]$. Also, because u_λ satisfies the equation $\mathcal{L}u = \lambda u$ for all $x \in [a, b]$, it follows that u_λ satisfies it for all $x \in [x_1, x_2] \subset [a, b]$. Hence $u_\lambda|_{[x_1, x_2]}$ is an eigenfunction of $\mathcal{L}$ on $[x_1, x_2]$ with eigenvalue λ . Because 1 is always an eigenfunction with

eigenvalues 0 (since $\mathcal{L}(1) = (p \cdot 0)' = 0$), and eigenfunctions with distinct eigenvalues are orthogonal we have that

$$\int_{x_1}^{x_2} u_\lambda(x) \cdot 1 \, dx = 0.$$

Now, suppose for contradiction that u_λ has no zero between (x_1, x_2) . By the intermediate value theorem u_λ must be either strictly positive or negative on the interval. However, if u_λ was strictly positive or negative then

$$\int_{x_1}^{x_2} u_\lambda(x) \, dx \neq 0$$

since $u_\lambda(x)$ is nontrivial. This is a contradiction to our earlier statement and so u_λ must have at least one zero in (x_1, x_2) . $\square$

4. Set $\mathcal{L}u(x) = ((1+x)u'(x))'$, so that $V_\lambda = \frac{1}{4} \frac{1}{(1+x)^2} - \frac{\lambda}{1+x}$ for Dirichlet boundary conditions on $[x_1, x_2]$. Assume $x_1, x_2 > 0$. Use the theorem to provide an upper and lower bound for λ_2 .

Solution:

Proof. Since $x_1, x_2 > 0$, the function $1/(1+x)$ is strictly decreasing, so V_λ is also strictly decreasing on $[x_1, x_2]$. Therefore

$$\begin{aligned} m^2 = V_\lambda(x_2) &= \frac{1}{4(1+x_2)^2} - \frac{\lambda^2}{1+x_2} \\ M^2 = V_\lambda(x_1) &= \frac{1}{4(1+x_1)^2} - \frac{\lambda_2}{1+x_1}. \end{aligned}$$

It follows from the theorem that

$$\frac{\pi}{M} \leq x_2 - x_1 \leq \frac{\pi}{m} \Rightarrow \frac{\pi^2}{M^2} \leq (x_2 - x_1)^2 \leq \frac{\pi^2}{m^2} \Rightarrow m^2 \leq \frac{\pi^2}{(x_2 - x_1)^2} \leq M^2.$$

Separating the inequalities and simplifying we see

$$\begin{aligned} m^2 \leq \frac{\pi^2}{(x_2 - x_1)^2} &\Rightarrow \frac{1}{4(1+x_2)^2} - \frac{\lambda_2}{1+x_2} \leq \frac{\pi^2}{(x_2 - x_1)^2} \\ &\Rightarrow \frac{1}{4}(1+x_2) - \lambda_2 \leq \frac{\pi^2(1+x_2)}{(x_2 - x_1)^2} \\ &\Rightarrow \lambda_2 \geq \frac{1}{4(1+x_2)} - \frac{\pi^2(1+x_2)}{(x_2 - x_1)^2} \end{aligned}$$

and

$$\begin{aligned} M^2 \geq \frac{\pi^2}{(x_2 - x_1)^2} &\Rightarrow \frac{1}{4(1+x_1)^2} - \frac{\lambda_2}{1+x_1} \geq \frac{\pi^2}{(x_2 - x_1)^2} \\ &\Rightarrow \frac{1}{4(1+x_1)} - \lambda_2 \geq \frac{\pi^2(1+x_1)}{(x_2 - x_1)^2} \\ &\Rightarrow \lambda_2 \leq \frac{1}{4(1+x_1)} - \frac{\pi^2(1+x_1)}{(x_2 - x_1)^2}. \end{aligned}$$

Thus, we can bound λ_2 by

$$\frac{1}{4(1+x_2)} - \frac{\pi^2(1+x_2)}{(x_2 - x_1)^2} \leq \lambda_2 \leq \frac{1}{4(1+x_1)} - \frac{\pi^2(1+x_1)}{(x_2 - x_1)^2}$$

$\square$